

DRAFT PROVISIONAL PATENT APPLICATIONS

Series A: the twenty-six JoinAQAL sister combinations. Every combination in this series is implemented and operating in the JoinAQAL codebase — software reduction to practice.

STATUS: DRAFT — NOT FILED. No application numbers exist; nothing herein is patent-pending. Per company records; confirm status with patent counsel. Prepared with AI assistance; not legal advice; attorney review required before filing.

Inventor: Samuel Andrew Russell V — Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-01

Title of Invention: Consensus-Driven Coaching Synthesis

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Eight-Model Consensus Scoring + Evidence-Mapped Protocol Coaching to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Eight-Model Consensus Scoring Engine:** a multi-model scoring architecture that improves measurement reliability over any single model by combining a trimmed-mean outlier guard with per-model, per-dimension calibration weights accumulated from real scoring history — a specific, non-generic weighting data structure, not a mental process.

- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Combinator. The combination executes as the consensusDrivenCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Eight-Model Consensus Scoring Engine:** Panel of eight models; trimmed-mean guard active whenever four or more models contribute (single highest and single lowest raw scores excluded per line); calibration weight $w = 1/(\text{meanAbsErr} + 0.02)$ with an error floor preventing weight explosion; equal weight 1.0 until a model accumulates five observations on a line; confidence = $0.5 \times \text{mean model confidence} + 0.5 \times \text{inter-model agreement}$.

- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.

Measured Performance

- Full 8-model x 32-line consensus combine: 0.11 ms (9,440 combines/sec).

- Panel fan-out over the compute fabric: 7.9x parallel speedup over serial execution (software backend; hardware co-processor figures are not claimed).

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Single-model AI scoring and unweighted ensemble averaging: neither maintains per-model, per-dimension calibration history with a cold-start guarantee and a retained trim guard.
- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein panel fan-out executes over a typed parallel compute fabric with bounded concurrency, a hardware co-processor backend being substitutable behind the same interface without alteration of the method.
6. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.
7. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
8. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
9. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
10. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
11. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-02

Title of Invention: Transparent Integral Framework

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Integral Four-Quadrant Mapper, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Integral Four-Quadrant + Transparency and Tamper-Evident Audit to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Integral Four-Quadrant Mapper:** a deterministic structural mapping that converts a 32-dimension measurement into a four-quadrant integral representation without human interpretive variance.
- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself,

verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* `server/patents/combinations/index.ts` (integral engine).

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* `server/patents/ledger.ts`; `/api/ledger/verify`; `/api/ledger/head`.

Combinator. The combination executes as the transparentIntegral combinator (`server/patents/combinations/index.ts`), registered in the sister-combination registry (`server/patents/combinations/registry.ts`) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Integral Four-Quadrant Mapper:** Deterministic quadrant assignment of all 32 lines across interior/exterior x individual/collective; the mapping table is versioned with the rubric.

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (`scripts/patentBenchmarks.test.ts`; `docs/PATENT_BENCHMARKS.md`) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Narrative integral-framework application: replaced by a deterministic, versioned mapping.

- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (b) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval $[0,1]$, a violating write being rejected as an error rather than stored.
5. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
6. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
7. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
8. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

9. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-03

Title of Invention: Voice-Based Weakness Identification

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Controlling-Weakness Identification Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Prosodic Voice-Feature Extraction + Controlling-Weakness Identification to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Prosodic Voice-Feature Extraction Engine:** an on-server digital-signal-processing pipeline whose energy-gated, lag-bounded pitch estimator eliminates the silent-frame false-pitch and subharmonic-bias failure modes of naive autocorrelation, verified by synthetic-tone tests — an improvement in the functioning of the audio-analysis computer itself.

- **Controlling-Weakness Identification Engine:** a constant-time deterministic diagnostic over a fixed-width vector that replaces unstructured narrative assessment with a reproducible, auditable computation.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Combinator. The combination executes as the voiceWeakness combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Prosodic Voice-Feature Extraction Engine:** Decoded mono PCM analyzed in fixed 50 ms pitch frames; cumulative-mean-normalized difference threshold 0.15 with descent to the local minimum; maximum candidate lag bounded to half the frame length; silent-frame energy gate (mean energy per sample below 1e-4 returns no pitch), eliminating false pitch in silence.

- **Controlling-Weakness Identification Engine:** Deterministic argmin over the fixed 32-dimension vector; constraint impact = clamp(profile mean minus minimum, 0..1); measured cost 0.11 microseconds per operation.

Measured Performance

- Full voice pipeline on 30 s of audio: 5,270 ms — 6x faster than real time.

- Pitch recovery on synthetic tones: 110 Hz -> 110.3 Hz (0.3% error); 220 Hz -> 219.2 Hz (0.4%); 330 Hz -> 333.3 Hz (1.0%).

- Controlling-weakness argmin over the 32-line vector: 0.11 microseconds per operation (8,745,295 ops/sec).

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Cloud speech-to-text vendors: prosody here is computed in-process on the operator's server under explicit on-page disclosure, with raw audio never transmitted to a third party.
- Composite/average scoring instruments: the claimed method elevates the minimum, not the mean, following limiting-factor theory (Liebig; O-ring), and quantifies constraint impact.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, and bounds the maximum candidate lag to half the analysis-frame length, preventing false pitch in silence and subharmonic bias, and wherein the raw audio never leaves the server for the claimed analysis.

5. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.
6. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.
7. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
8. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
9. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
10. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
11. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-04

Title of Invention: Voice-Transparent Consensus Assessment

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Prosodic Voice-Feature Extraction Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Eight-Model Consensus Scoring + Prosodic Voice-Feature Extraction + Transparency and Tamper-Evident Audit to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Eight-Model Consensus Scoring Engine:** a multi-model scoring architecture that improves measurement reliability over any single model by combining a trimmed-mean outlier guard with per-model, per-dimension calibration weights accumulated from real scoring history — a specific, non-generic weighting data structure, not a mental process.

- **Prosodic Voice-Feature Extraction Engine:** an on-server digital-signal-processing pipeline whose energy-gated, lag-bounded pitch estimator eliminates the silent-frame false-pitch and subharmonic-bias failure modes of naive autocorrelation, verified by synthetic-tone tests — an improvement in the functioning of the audio-analysis computer itself.

- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the voiceTransparentConsensus combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Eight-Model Consensus Scoring Engine:** Panel of eight models; trimmed-mean guard active whenever four or more models contribute (single highest and single lowest raw scores excluded per line); calibration weight $w =$

$1/(\text{meanAbsErr} + 0.02)$ with an error floor preventing weight explosion; equal weight 1.0 until a model accumulates five observations on a line; confidence = $0.5 \times \text{mean model confidence} + 0.5 \times \text{inter-model agreement}$.

- **Prosodic Voice-Feature Extraction Engine:** Decoded mono PCM analyzed in fixed 50 ms pitch frames; cumulative-mean-normalized difference threshold 0.15 with descent to the local minimum; maximum candidate lag bounded to half the frame length; silent-frame energy gate (mean energy per sample below $1e-4$ returns no pitch), eliminating false pitch in silence.

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Full 8-model x 32-line consensus combine: 0.11 ms (9,440 combines/sec).

- Panel fan-out over the compute fabric: 7.9x parallel speedup over serial execution (software backend; hardware co-processor figures are not claimed).

- Full voice pipeline on 30 s of audio: 5,270 ms — 6x faster than real time.

- Pitch recovery on synthetic tones: 110 Hz -> 110.3 Hz (0.3% error); 220 Hz -> 219.2 Hz (0.4%); 330 Hz -> 333.3 Hz (1.0%).

- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Single-model AI scoring and unweighted ensemble averaging: neither maintains per-model, per-dimension calibration history with a cold-start guarantee and a retained trim guard.

- Cloud speech-to-text vendors: prosody here is computed in-process on the operator's server under explicit on-page disclosure, with raw audio never transmitted to a third party.

- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is

disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein panel fan-out executes over a typed parallel compute fabric with bounded concurrency, a hardware co-processor backend being substitutable behind the same interface without alteration of the method.
6. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, and bounds the maximum candidate lag to half the analysis-frame length, preventing false pitch in silence and subharmonic bias, and wherein the raw audio never leaves the server for the claimed analysis.
7. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval $[0,1]$, a violating write being rejected as an error rather than stored.
8. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
9. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
10. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

11. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
12. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-05

Title of Invention: Staged Weakness Identification Engine

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Developmental Stage-Band Estimator, Controlling-Weakness Identification Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Developmental Stage-Band + Controlling-Weakness Identification to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Developmental Stage-Band Estimator:** versioned-rubric banding that makes developmental staging reproducible: the same inputs under the same frozen rubric version always produce the same band.
- **Controlling-Weakness Identification Engine:** a constant-time deterministic diagnostic over a fixed-width vector that replaces unstructured narrative assessment with a reproducible, auditable computation.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* `server/platform/stageFramework.ts`; `references/NineStageRubric.pdf`.

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* `server/scoring/controllingWeakness.ts`.

Combinator. The combination executes as the stagedWeakness combinator (`server/patents/combinations/index.ts`), registered in the sister-combination registry (`server/patents/combinations/registry.ts`) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Developmental Stage-Band Estimator:** Stage banding computed under a fixed, versioned nine-stage rubric frozen per norming snapshot, so identical inputs reproduce identical bands at any later date.
- **Controlling-Weakness Identification Engine:** Deterministic argmin over the fixed 32-dimension vector; constraint impact = $\text{clamp}(\text{profile mean minus minimum}, 0..1)$; measured cost 0.11 microseconds per operation.

Measured Performance

- Controlling-weakness argmin over the 32-line vector: 0.11 microseconds per operation (8,745,295 ops/sec).

All figures measured by the repository's benchmark harness (`scripts/patentBenchmarks.test.ts`; `docs/PATENT_BENCHMARKS.md`) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Practitioner-scored developmental staging: replaced by a frozen, versioned rubric computation that is reproducible and auditable.
- Composite/average scoring instruments: the claimed method elevates the minimum, not the mean, following limiting-factor theory (Liebig; O-ring), and quantifies constraint impact.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (b) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.
5. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval $[0,1]$, a violating write being rejected as an error rather than stored.
6. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
7. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
8. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
9. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
10. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-06

Title of Invention: Coaching Transparency Framework

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Evidence-Mapped Protocol Coaching Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Evidence-Mapped Protocol Coaching + Transparency and Tamper-Evident Audit to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.

- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* `shared/therapyLineMap.ts`; `server/patents/weaknessInterventions.ts`.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* `server/patents/ledger.ts`; `/api/ledger/verify`; `/api/ledger/head`.

Combinator. The combination executes as the `coachingTransparency` combinator (`server/patents/combinations/index.ts`), registered in the sister-combination registry (`server/patents/combinations/registry.ts`) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (`scripts/patentBenchmarks.test.ts`; `docs/PATENT_BENCHMARKS.md`) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.
- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (b) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.
5. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public

changelog enumerates every version.

6. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.

7. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

8. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

9. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**

- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-07

Title of Invention: Voice-Driven Weakness Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Prosodic Voice-Feature Extraction + Controlling-Weakness Identification + Evidence-Mapped Protocol Coaching to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Prosodic Voice-Feature Extraction Engine:** an on-server digital-signal-processing pipeline whose energy-gated, lag-bounded pitch estimator eliminates the silent-frame false-pitch and subharmonic-bias failure modes of naive autocorrelation, verified by synthetic-tone tests — an improvement in the functioning of the audio-analysis computer

itself.

- **Controlling-Weakness Identification Engine:** a constant-time deterministic diagnostic over a fixed-width vector that replaces unstructured narrative assessment with a reproducible, auditable computation.

- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Combinator. The combination executes as the voiceWeaknessCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Prosodic Voice-Feature Extraction Engine:** Decoded mono PCM analyzed in fixed 50 ms pitch frames; cumulative-mean-normalized difference threshold 0.15 with descent to the local minimum; maximum candidate lag bounded to half the frame length; silent-frame energy gate (mean energy per sample below 1e-4 returns no pitch), eliminating false pitch in silence.

- **Controlling-Weakness Identification Engine:** Deterministic argmin over the fixed 32-dimension vector; constraint impact = clamp(profile mean minus minimum, 0..1); measured cost 0.11 microseconds per operation.

- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.

Measured Performance

- Full voice pipeline on 30 s of audio: 5,270 ms — 6x faster than real time.
- Pitch recovery on synthetic tones: 110 Hz -> 110.3 Hz (0.3% error); 220 Hz -> 219.2 Hz (0.4%); 330 Hz -> 333.3 Hz (1.0%).
- Controlling-weakness argmin over the 32-line vector: 0.11 microseconds per operation (8,745,295 ops/sec).

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Cloud speech-to-text vendors: prosody here is computed in-process on the operator's server under explicit on-page disclosure, with raw audio never transmitted to a third party.
- Composite/average scoring instruments: the claimed method elevates the minimum, not the mean, following limiting-factor theory (Liebig; O-ring), and quantifies constraint impact.
- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) deterministically identifying the

single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (c) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, and bounds the maximum candidate lag to half the analysis-frame length, preventing false pitch in silence and subharmonic bias, and wherein the raw audio never leaves the server for the claimed analysis.

5. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.

6. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.

7. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.

8. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.

9. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

10. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

11. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-08

Title of Invention: Transparent Staged Consensus Assessment

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Developmental Stage-Band Estimator, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Eight-Model Consensus Scoring + Developmental Stage-Band + Transparency and Tamper-Evident Audit to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Eight-Model Consensus Scoring Engine:** a multi-model scoring architecture that improves measurement reliability over any single model by combining a trimmed-mean outlier guard with per-model, per-dimension calibration weights accumulated from real scoring history — a specific, non-generic weighting data structure, not a mental process.

- **Developmental Stage-Band Estimator:** versioned-rubric banding that makes developmental staging reproducible: the same inputs under the same frozen rubric version always produce the same band.

- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the transparentStagedConsensus combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Eight-Model Consensus Scoring Engine:** Panel of eight models; trimmed-mean guard active whenever four or more models contribute (single highest and single lowest raw scores excluded per line); calibration weight $w = 1/(\text{meanAbsErr} + 0.02)$ with an error floor preventing weight explosion; equal weight 1.0 until a model accumulates five observations on a line; confidence = $0.5 \times \text{mean model confidence} + 0.5 \times \text{inter-model agreement}$.

- **Developmental Stage-Band Estimator:** Stage banding computed under a fixed, versioned nine-stage rubric frozen per norming snapshot, so identical inputs reproduce identical bands at any later date.

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Full 8-model x 32-line consensus combine: 0.11 ms (9,440 combines/sec).
- Panel fan-out over the compute fabric: 7.9x parallel speedup over serial execution (software backend; hardware co-processor figures are not claimed).
- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Single-model AI scoring and unweighted ensemble averaging: neither maintains per-model, per-dimension calibration history with a cold-start guarantee and a retained trim guard.
- Practitioner-scored developmental staging: replaced by a frozen, versioned rubric computation that is reproducible and auditable.
- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQUAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum

sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.

5. The method of claim 1, wherein panel fan-out executes over a typed parallel compute fabric with bounded concurrency, a hardware co-processor backend being substitutable behind the same interface without alteration of the method.

6. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.

7. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.

8. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.

9. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

10. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

11. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-09

Title of Invention: Integral Voice Consensus

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Prosodic Voice-Feature Extraction Engine, Integral Four-Quadrant Mapper. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Eight-Model Consensus Scoring + Prosodic Voice-Feature Extraction + Integral Four-Quadrant to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Eight-Model Consensus Scoring Engine:** a multi-model scoring architecture that improves measurement reliability over any single model by combining a trimmed-mean outlier guard with per-model, per-dimension calibration weights accumulated from real scoring history — a specific, non-generic weighting data structure, not a mental process.

- **Prosodic Voice-Feature Extraction Engine:** an on-server digital-signal-processing pipeline whose energy-gated, lag-bounded pitch estimator eliminates the silent-frame false-pitch and subharmonic-bias failure modes of naive autocorrelation, verified by synthetic-tone tests — an improvement in the functioning of the audio-analysis computer itself.

- **Integral Four-Quadrant Mapper:** a deterministic structural mapping that converts a 32-dimension measurement into a four-quadrant integral representation without human interpretive variance.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* server/patents/combinations/index.ts (integral engine).

Combinator. The combination executes as the integralVoiceConsensus combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Eight-Model Consensus Scoring Engine:** Panel of eight models; trimmed-mean guard active whenever four or more models contribute (single highest and single lowest raw scores excluded per line); calibration weight $w = 1/(\text{meanAbsErr} + 0.02)$ with an error floor preventing weight explosion; equal weight 1.0 until a model accumulates five observations on a line; confidence = $0.5 \times \text{mean model confidence} + 0.5 \times \text{inter-model agreement}$.

- **Prosodic Voice-Feature Extraction Engine:** Decoded mono PCM analyzed in fixed 50 ms pitch frames; cumulative-mean-normalized difference threshold 0.15 with descent to the local minimum; maximum candidate lag

bounded to half the frame length; silent-frame energy gate (mean energy per sample below $1e-4$ returns no pitch), eliminating false pitch in silence.

- **Integral Four-Quadrant Mapper:** Deterministic quadrant assignment of all 32 lines across interior/exterior x individual/collective; the mapping table is versioned with the rubric.

Measured Performance

- Full 8-model x 32-line consensus combine: 0.11 ms (9,440 combines/sec).
- Panel fan-out over the compute fabric: 7.9x parallel speedup over serial execution (software backend; hardware co-processor figures are not claimed).
- Full voice pipeline on 30 s of audio: 5,270 ms — 6x faster than real time.
- Pitch recovery on synthetic tones: 110 Hz -> 110.3 Hz (0.3% error); 220 Hz -> 219.2 Hz (0.4%); 330 Hz -> 333.3 Hz (1.0%).

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Single-model AI scoring and unweighted ensemble averaging: neither maintains per-model, per-dimension calibration history with a cold-start guarantee and a retained trim guard.
- Cloud speech-to-text vendors: prosody here is computed in-process on the operator's server under explicit on-page disclosure, with raw audio never transmitted to a third party.
- Narrative integral-framework application: replaced by a deterministic, versioned mapping.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQUAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining

the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (c) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.

5. The method of claim 1, wherein panel fan-out executes over a typed parallel compute fabric with bounded concurrency, a hardware co-processor backend being substitutable behind the same interface without alteration of the method.

6. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, and bounds the maximum candidate lag to half the analysis-frame length, preventing false pitch in silence and subharmonic bias, and wherein the raw audio never leaves the server for the claimed analysis.

7. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.

8. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.

9. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.

10. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

11. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

12. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-10

Title of Invention: Staged Weakness Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Developmental Stage-Band Estimator, Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Developmental Stage-Band + Controlling-Weakness Identification + Evidence-Mapped Protocol Coaching to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Developmental Stage-Band Estimator:** versioned-rubric banding that makes developmental staging reproducible: the same inputs under the same frozen rubric version always produce the same band.
- **Controlling-Weakness Identification Engine:** a constant-time deterministic diagnostic over a fixed-width vector that replaces unstructured narrative assessment with a reproducible, auditable computation.

- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Combinator. The combination executes as the stagedWeaknessCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Developmental Stage-Band Estimator:** Stage banding computed under a fixed, versioned nine-stage rubric frozen per norming snapshot, so identical inputs reproduce identical bands at any later date.

- **Controlling-Weakness Identification Engine:** Deterministic argmin over the fixed 32-dimension vector; constraint impact = clamp(profile mean minus minimum, 0..1); measured cost 0.11 microseconds per operation.

- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.

Measured Performance

- Controlling-weakness argmin over the 32-line vector: 0.11 microseconds per operation (8,745,295 ops/sec).

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment

numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Practitioner-scored developmental staging: replaced by a frozen, versioned rubric computation that is reproducible and auditable.
- Composite/average scoring instruments: the claimed method elevates the minimum, not the mean, following limiting-factor theory (Liebig; O-ring), and quantifies constraint impact.
- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (b) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (c) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.
5. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.
6. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
7. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
8. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
9. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
10. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-11

Title of Invention: Transparent Integral Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Integral Four-Quadrant Mapper, Evidence-Mapped Protocol Coaching Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Integral Four-Quadrant + Evidence-Mapped Protocol Coaching + Transparency and Tamper-Evident Audit to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Integral Four-Quadrant Mapper:** a deterministic structural mapping that converts a 32-dimension measurement into a four-quadrant integral representation without human interpretive variance.

- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.

- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* server/patents/combinations/index.ts (integral engine).

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the transparentIntegralCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Integral Four-Quadrant Mapper:** Deterministic quadrant assignment of all 32 lines across interior/exterior x individual/collective; the mapping table is versioned with the rubric.

- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Narrative integral-framework application: replaced by a deterministic, versioned mapping.
- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.
- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (b) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.
5. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
6. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
7. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
8. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
9. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-12

Title of Invention: Voice-Driven Staged Weakness Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Developmental Stage-Band Estimator, Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Prosodic Voice-Feature Extraction + Developmental Stage-Band + Controlling-Weakness Identification + Evidence-Mapped Protocol Coaching to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Prosodic Voice-Feature Extraction Engine:** an on-server digital-signal-processing pipeline whose energy-gated, lag-bounded pitch estimator eliminates the silent-frame false-pitch and subharmonic-bias failure modes of naive

autocorrelation, verified by synthetic-tone tests — an improvement in the functioning of the audio-analysis computer itself.

- **Developmental Stage-Band Estimator:** versioned-rubric banding that makes developmental staging reproducible: the same inputs under the same frozen rubric version always produce the same band.

- **Controlling-Weakness Identification Engine:** a constant-time deterministic diagnostic over a fixed-width vector that replaces unstructured narrative assessment with a reproducible, auditable computation.

- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Combinator. The combination executes as the voiceStagedWeaknessCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Prosodic Voice-Feature Extraction Engine:** Decoded mono PCM analyzed in fixed 50 ms pitch frames; cumulative-mean-normalized difference threshold 0.15 with descent to the local minimum; maximum candidate lag bounded to half the frame length; silent-frame energy gate (mean energy per sample below $1e-4$ returns no pitch), eliminating false pitch in silence.
- **Developmental Stage-Band Estimator:** Stage banding computed under a fixed, versioned nine-stage rubric frozen per norming snapshot, so identical inputs reproduce identical bands at any later date.
- **Controlling-Weakness Identification Engine:** Deterministic argmin over the fixed 32-dimension vector; constraint impact = clamp(profile mean minus minimum, 0..1); measured cost 0.11 microseconds per operation.
- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.

Measured Performance

- Full voice pipeline on 30 s of audio: 5,270 ms — 6x faster than real time.
- Pitch recovery on synthetic tones: 110 Hz -> 110.3 Hz (0.3% error); 220 Hz -> 219.2 Hz (0.4%); 330 Hz -> 333.3 Hz (1.0%).
- Controlling-weakness argmin over the 32-line vector: 0.11 microseconds per operation (8,745,295 ops/sec).

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Cloud speech-to-text vendors: prosody here is computed in-process on the operator's server under explicit on-page disclosure, with raw audio never transmitted to a third party.
- Practitioner-scored developmental staging: replaced by a frozen, versioned rubric computation that is reproducible and auditable.
- Composite/average scoring instruments: the claimed method elevates the minimum, not the mean, following limiting-factor theory (Liebig; O-ring), and quantifies constraint impact.
- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (c) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (d) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (e) combining the outputs of steps (a)–(d) into a single emergent result not produced by any component engine alone; and (f) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, and bounds the maximum candidate lag to half the analysis-frame length, preventing false pitch in silence and subharmonic bias, and wherein the raw audio never leaves the server for the claimed analysis.
5. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.
6. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.
7. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
8. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
9. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

10. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

11. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-13

Title of Invention: Transparent Integral Consensus Assessment

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Integral Four-Quadrant Mapper, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Eight-Model Consensus Scoring + Integral Four-Quadrant + Transparency and Tamper-Evident Audit to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Eight-Model Consensus Scoring Engine:** a multi-model scoring architecture that improves measurement reliability over any single model by combining a trimmed-mean outlier guard with per-model, per-dimension calibration weights accumulated from real scoring history — a specific, non-generic weighting data structure, not a mental process.

- **Integral Four-Quadrant Mapper:** a deterministic structural mapping that converts a 32-dimension measurement into a four-quadrant integral representation without human interpretive variance.

- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* server/patents/combinations/index.ts (integral engine).

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the transparentIntegralConsensus combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Eight-Model Consensus Scoring Engine:** Panel of eight models; trimmed-mean guard active whenever four or more models contribute (single highest and single lowest raw scores excluded per line); calibration weight $w = 1/(\text{meanAbsErr} + 0.02)$ with an error floor preventing weight explosion; equal weight 1.0 until a model accumulates five observations on a line; confidence = $0.5 \times \text{mean model confidence} + 0.5 \times \text{inter-model agreement}$.

- **Integral Four-Quadrant Mapper:** Deterministic quadrant assignment of all 32 lines across interior/exterior x individual/collective; the mapping table is versioned with the rubric.

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Full 8-model x 32-line consensus combine: 0.11 ms (9,440 combines/sec).
- Panel fan-out over the compute fabric: 7.9x parallel speedup over serial execution (software backend; hardware co-processor figures are not claimed).
- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Single-model AI scoring and unweighted ensemble averaging: neither maintains per-model, per-dimension calibration history with a cold-start guarantee and a retained trim guard.
- Narrative integral-framework application: replaced by a deterministic, versioned mapping.
- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) mapping the 32-line

profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.

5. The method of claim 1, wherein panel fan-out executes over a typed parallel compute fabric with bounded concurrency, a hardware co-processor backend being substitutable behind the same interface without alteration of the method.

6. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.

7. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.

8. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.

9. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

10. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

11. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-14

Title of Invention: Staged Integral Weakness Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Developmental Stage-Band Estimator, Integral Four-Quadrant Mapper, Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Developmental Stage-Band + Integral Four-Quadrant + Controlling-Weakness Identification + Evidence-Mapped Protocol Coaching to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Developmental Stage-Band Estimator:** versioned-rubric banding that makes developmental staging reproducible: the same inputs under the same frozen rubric version always produce the same band.

- **Integral Four-Quadrant Mapper:** a deterministic structural mapping that converts a 32-dimension measurement into a four-quadrant integral representation without human interpretive variance.
- **Controlling-Weakness Identification Engine:** a constant-time deterministic diagnostic over a fixed-width vector that replaces unstructured narrative assessment with a reproducible, auditable computation.
- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* server/patents/combinations/index.ts (integral engine).

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Combinator. The combination executes as the stagedIntegralWeaknessCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Developmental Stage-Band Estimator:** Stage banding computed under a fixed, versioned nine-stage rubric frozen per norming snapshot, so identical inputs reproduce identical bands at any later date.
- **Integral Four-Quadrant Mapper:** Deterministic quadrant assignment of all 32 lines across interior/exterior x individual/collective; the mapping table is versioned with the rubric.

- **Controlling-Weakness Identification Engine:** Deterministic argmin over the fixed 32-dimension vector; constraint impact = clamp(profile mean minus minimum, 0..1); measured cost 0.11 microseconds per operation.

- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.

Measured Performance

- Controlling-weakness argmin over the 32-line vector: 0.11 microseconds per operation (8,745,295 ops/sec).

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Practitioner-scored developmental staging: replaced by a frozen, versioned rubric computation that is reproducible and auditable.
- Narrative integral-framework application: replaced by a deterministic, versioned mapping.
- Composite/average scoring instruments: the claimed method elevates the minimum, not the mean, following limiting-factor theory (Liebig; O-ring), and quantifies constraint impact.
- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (b) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (c) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector

and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (d) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (e) combining the outputs of steps (a)–(d) into a single emergent result not produced by any component engine alone; and (f) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.

5. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.

6. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.

7. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.

8. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

9. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

10. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-15

Title of Invention: Consensus-Driven Integral Coaching with Transparency

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Integral Four-Quadrant Mapper, Evidence-Mapped Protocol Coaching Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Eight-Model Consensus Scoring + Integral Four-Quadrant + Evidence-Mapped Protocol Coaching + Transparency and Tamper-Evident Audit to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Eight-Model Consensus Scoring Engine:** a multi-model scoring architecture that improves measurement reliability over any single model by combining a trimmed-mean outlier guard with per-model, per-dimension calibration weights

accumulated from real scoring history — a specific, non-generic weighting data structure, not a mental process.

- **Integral Four-Quadrant Mapper:** a deterministic structural mapping that converts a 32-dimension measurement into a four-quadrant integral representation without human interpretive variance.

- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.

- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* server/patents/combinations/index.ts (integral engine).

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the consensusIntegralCoachingTransparency combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Eight-Model Consensus Scoring Engine:** Panel of eight models; trimmed-mean guard active whenever four or more models contribute (single highest and single lowest raw scores excluded per line); calibration weight $w = 1/(\text{meanAbsErr} + 0.02)$ with an error floor preventing weight explosion; equal weight 1.0 until a model accumulates five observations on a line; confidence = $0.5 \times \text{mean model confidence} + 0.5 \times \text{inter-model agreement}$.
- **Integral Four-Quadrant Mapper:** Deterministic quadrant assignment of all 32 lines across interior/exterior x individual/collective; the mapping table is versioned with the rubric.
- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.
- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Full 8-model x 32-line consensus combine: 0.11 ms (9,440 combines/sec).
- Panel fan-out over the compute fabric: 7.9x parallel speedup over serial execution (software backend; hardware co-processor figures are not claimed).
- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Single-model AI scoring and unweighted ensemble averaging: neither maintains per-model, per-dimension calibration history with a cold-start guarantee and a retained trim guard.
- Narrative integral-framework application: replaced by a deterministic, versioned mapping.
- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.
- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (c) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (d) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (e) combining the outputs of steps (a)–(d) into a single emergent result not produced by any component engine alone; and (f) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein panel fan-out executes over a typed parallel compute fabric with bounded concurrency, a hardware co-processor backend being substitutable behind the same interface without alteration of the method.
6. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.
7. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
8. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
9. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

10. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

11. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-16

Title of Invention: Voice-Driven Staged Integral Weakness Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

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Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Developmental Stage-Band Estimator, Integral Four-Quadrant Mapper, Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Prosodic Voice-Feature Extraction + Developmental Stage-Band + Integral Four-Quadrant + Controlling-Weakness Identification + Evidence-Mapped Protocol Coaching to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Prosodic Voice-Feature Extraction Engine:** an on-server digital-signal-processing pipeline whose energy-gated, lag-bounded pitch estimator eliminates the silent-frame false-pitch and subharmonic-bias failure modes of naive

autocorrelation, verified by synthetic-tone tests — an improvement in the functioning of the audio-analysis computer itself.

- **Developmental Stage-Band Estimator:** versioned-rubric banding that makes developmental staging reproducible: the same inputs under the same frozen rubric version always produce the same band.

- **Integral Four-Quadrant Mapper:** a deterministic structural mapping that converts a 32-dimension measurement into a four-quadrant integral representation without human interpretive variance.

- **Controlling-Weakness Identification Engine:** a constant-time deterministic diagnostic over a fixed-width vector that replaces unstructured narrative assessment with a reproducible, auditable computation.

- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* server/patents/combinations/index.ts (integral engine).

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Combinator. The combination executes as the voiceStagedIntegralWeaknessCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by

construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Prosodic Voice-Feature Extraction Engine:** Decoded mono PCM analyzed in fixed 50 ms pitch frames; cumulative-mean-normalized difference threshold 0.15 with descent to the local minimum; maximum candidate lag bounded to half the frame length; silent-frame energy gate (mean energy per sample below $1e-4$ returns no pitch), eliminating false pitch in silence.
- **Developmental Stage-Band Estimator:** Stage banding computed under a fixed, versioned nine-stage rubric frozen per norming snapshot, so identical inputs reproduce identical bands at any later date.
- **Integral Four-Quadrant Mapper:** Deterministic quadrant assignment of all 32 lines across interior/exterior x individual/collective; the mapping table is versioned with the rubric.
- **Controlling-Weakness Identification Engine:** Deterministic argmin over the fixed 32-dimension vector; constraint impact = clamp(profile mean minus minimum, 0..1); measured cost 0.11 microseconds per operation.
- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.

Measured Performance

- Full voice pipeline on 30 s of audio: 5,270 ms — 6x faster than real time.
- Pitch recovery on synthetic tones: 110 Hz -> 110.3 Hz (0.3% error); 220 Hz -> 219.2 Hz (0.4%); 330 Hz -> 333.3 Hz (1.0%).
- Controlling-weakness argmin over the 32-line vector: 0.11 microseconds per operation (8,745,295 ops/sec).

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Cloud speech-to-text vendors: prosody here is computed in-process on the operator's server under explicit on-page disclosure, with raw audio never transmitted to a third party.
- Practitioner-scored developmental staging: replaced by a frozen, versioned rubric computation that is reproducible and auditable.
- Narrative integral-framework application: replaced by a deterministic, versioned mapping.
- Composite/average scoring instruments: the claimed method elevates the minimum, not the mean, following limiting-factor theory (Liebig; O-ring), and quantifies constraint impact.
- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (c) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (d) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (e) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (f) combining the outputs of steps (a)–(e) into a single emergent result not produced by any component engine alone; and (g) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, and bounds the maximum candidate lag to half the analysis-frame length, preventing false pitch in silence and subharmonic bias, and wherein the raw audio never leaves the server for the claimed analysis.
5. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.
6. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.

7. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
8. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
9. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
10. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
11. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-17

Title of Invention: Full-Spectrum Integral AI Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

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This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Prosodic Voice-Feature Extraction Engine, Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine, Developmental Stage-Band Estimator, Integral Four-Quadrant Mapper, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Eight-Model Consensus Scoring + Prosodic Voice-Feature Extraction + Controlling-Weakness Identification + Evidence-Mapped Protocol Coaching + Developmental Stage-Band + Integral Four-Quadrant + Transparency and Tamper-Evident Audit to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Eight-Model Consensus Scoring Engine:** a multi-model scoring architecture that improves measurement reliability over any single model by combining a trimmed-mean outlier guard with per-model, per-dimension calibration weights accumulated from real scoring history — a specific, non-generic weighting data structure, not a mental process.
- **Prosodic Voice-Feature Extraction Engine:** an on-server digital-signal-processing pipeline whose energy-gated, lag-bounded pitch estimator eliminates the silent-frame false-pitch and subharmonic-bias failure modes of naive autocorrelation, verified by synthetic-tone tests — an improvement in the functioning of the audio-analysis computer itself.
- **Controlling-Weakness Identification Engine:** a constant-time deterministic diagnostic over a fixed-width vector that replaces unstructured narrative assessment with a reproducible, auditable computation.
- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.
- **Developmental Stage-Band Estimator:** versioned-rubric banding that makes developmental staging reproducible: the same inputs under the same frozen rubric version always produce the same band.
- **Integral Four-Quadrant Mapper:** a deterministic structural mapping that converts a 32-dimension measurement into a four-quadrant integral representation without human interpretive variance.
- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* server/patents/combinations/index.ts (integral engine).

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the fullSpectrumIntegralCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Eight-Model Consensus Scoring Engine:** Panel of eight models; trimmed-mean guard active whenever four or more models contribute (single highest and single lowest raw scores excluded per line); calibration weight $w = 1/(\text{meanAbsErr} + 0.02)$ with an error floor preventing weight explosion; equal weight 1.0 until a model accumulates five observations on a line; confidence = $0.5 \times \text{mean model confidence} + 0.5 \times \text{inter-model agreement}$.

- **Prosodic Voice-Feature Extraction Engine:** Decoded mono PCM analyzed in fixed 50 ms pitch frames; cumulative-mean-normalized difference threshold 0.15 with descent to the local minimum; maximum candidate lag bounded to half the frame length; silent-frame energy gate (mean energy per sample below $1e-4$ returns no pitch), eliminating false pitch in silence.

- **Controlling-Weakness Identification Engine:** Deterministic argmin over the fixed 32-dimension vector; constraint impact = $\text{clamp}(\text{profile mean minus minimum}, 0..1)$; measured cost 0.11 microseconds per operation.

- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.

- **Developmental Stage-Band Estimator:** Stage banding computed under a fixed, versioned nine-stage rubric frozen per norming snapshot, so identical inputs reproduce identical bands at any later date.

- **Integral Four-Quadrant Mapper:** Deterministic quadrant assignment of all 32 lines across interior/exterior x individual/collective; the mapping table is versioned with the rubric.

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Full 8-model x 32-line consensus combine: 0.11 ms (9,440 combines/sec).
- Panel fan-out over the compute fabric: 7.9x parallel speedup over serial execution (software backend; hardware co-processor figures are not claimed).
- Full voice pipeline on 30 s of audio: 5,270 ms — 6x faster than real time.
- Pitch recovery on synthetic tones: 110 Hz -> 110.3 Hz (0.3% error); 220 Hz -> 219.2 Hz (0.4%); 330 Hz -> 333.3 Hz (1.0%).
- Controlling-weakness argmin over the 32-line vector: 0.11 microseconds per operation (8,745,295 ops/sec).
- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Single-model AI scoring and unweighted ensemble averaging: neither maintains per-model, per-dimension calibration history with a cold-start guarantee and a retained trim guard.
- Cloud speech-to-text vendors: prosody here is computed in-process on the operator's server under explicit on-page disclosure, with raw audio never transmitted to a third party.
- Composite/average scoring instruments: the claimed method elevates the minimum, not the mean, following limiting-factor theory (Liebig; O-ring), and quantifies constraint impact.
- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.
- Practitioner-scored developmental staging: replaced by a frozen, versioned rubric computation that is reproducible and auditable.
- Narrative integral-framework application: replaced by a deterministic, versioned mapping.
- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third

parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (c) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (d) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (e) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (f) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (g) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (h) combining the outputs of steps (a)–(g) into a single emergent result not produced by any component engine alone; and (i) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein panel fan-out executes over a typed parallel compute fabric with bounded concurrency, a hardware co-processor backend being substitutable behind the same interface without alteration of the method.
6. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, and bounds the maximum candidate lag to half the analysis-frame length, preventing false pitch in silence and subharmonic bias, and wherein the raw audio never leaves the server for the claimed analysis.

7. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.
8. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.
9. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
10. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
11. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
12. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
13. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-18

Title of Invention: Adaptive Voice Consensus

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Eight-Model Consensus Scoring Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Prosodic Voice-Feature Extraction + Eight-Model Consensus Scoring to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Prosodic Voice-Feature Extraction Engine:** an on-server digital-signal-processing pipeline whose energy-gated, lag-bounded pitch estimator eliminates the silent-frame false-pitch and subharmonic-bias failure modes of naive autocorrelation, verified by synthetic-tone tests — an improvement in the functioning of the audio-analysis computer itself.

- **Eight-Model Consensus Scoring Engine:** a multi-model scoring architecture that improves measurement reliability over any single model by combining a trimmed-mean outlier guard with per-model, per-dimension calibration weights accumulated from real scoring history — a specific, non-generic weighting data structure, not a mental process.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Combinator. The combination executes as the adaptiveVoiceConsensus combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Prosodic Voice-Feature Extraction Engine:** Decoded mono PCM analyzed in fixed 50 ms pitch frames; cumulative-mean-normalized difference threshold 0.15 with descent to the local minimum; maximum candidate lag bounded to half the frame length; silent-frame energy gate (mean energy per sample below 1e-4 returns no pitch), eliminating false pitch in silence.

- **Eight-Model Consensus Scoring Engine:** Panel of eight models; trimmed-mean guard active whenever four or more models contribute (single highest and single lowest raw scores excluded per line); calibration weight $w = 1/(\text{meanAbsErr} + 0.02)$ with an error floor preventing weight explosion; equal weight 1.0 until a model accumulates five observations on a line; confidence = $0.5 \times \text{mean model confidence} + 0.5 \times \text{inter-model agreement}$.

Measured Performance

- Full voice pipeline on 30 s of audio: 5,270 ms — 6x faster than real time.

- Pitch recovery on synthetic tones: 110 Hz -> 110.3 Hz (0.3% error); 220 Hz -> 219.2 Hz (0.4%); 330 Hz -> 333.3 Hz (1.0%).

- Full 8-model x 32-line consensus combine: 0.11 ms (9,440 combines/sec).

- Panel fan-out over the compute fabric: 7.9x parallel speedup over serial execution (software backend; hardware co-processor figures are not claimed).

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Cloud speech-to-text vendors: prosody here is computed in-process on the operator's server under explicit on-page disclosure, with raw audio never transmitted to a third party.

- Single-model AI scoring and unweighted ensemble averaging: neither maintains per-model, per-dimension calibration history with a cold-start guarantee and a retained trim guard.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQUAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only

hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein panel fan-out executes over a typed parallel compute fabric with bounded concurrency, a hardware co-processor backend being substitutable behind the same interface without alteration of the method.
6. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, and bounds the maximum candidate lag to half the analysis-frame length, preventing false pitch in silence and subharmonic bias, and wherein the raw audio never leaves the server for the claimed analysis.
7. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.
8. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
9. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
10. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
11. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
12. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-19

Title of Invention: Transparent Voice Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Evidence-Mapped Protocol Coaching Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Prosodic Voice-Feature Extraction + Evidence-Mapped Protocol Coaching + Transparency and Tamper-Evident Audit to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Prosodic Voice-Feature Extraction Engine:** an on-server digital-signal-processing pipeline whose energy-gated, lag-bounded pitch estimator eliminates the silent-frame false-pitch and subharmonic-bias failure modes of naive autocorrelation, verified by synthetic-tone tests — an improvement in the functioning of the audio-analysis computer

itself.

- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.

- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the transparentVoiceCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Prosodic Voice-Feature Extraction Engine:** Decoded mono PCM analyzed in fixed 50 ms pitch frames; cumulative-mean-normalized difference threshold 0.15 with descent to the local minimum; maximum candidate lag bounded to half the frame length; silent-frame energy gate (mean energy per sample below 1e-4 returns no pitch), eliminating false pitch in silence.

- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Full voice pipeline on 30 s of audio: 5,270 ms — 6x faster than real time.
- Pitch recovery on synthetic tones: 110 Hz -> 110.3 Hz (0.3% error); 220 Hz -> 219.2 Hz (0.4%); 330 Hz -> 333.3 Hz (1.0%).
- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Cloud speech-to-text vendors: prosody here is computed in-process on the operator's server under explicit on-page disclosure, with raw audio never transmitted to a third party.
- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.
- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) extracting, in-process on the

server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, and bounds the maximum candidate lag to half the analysis-frame length, preventing false pitch in silence and subharmonic bias, and wherein the raw audio never leaves the server for the claimed analysis.

5. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.

6. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.

7. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.

8. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

9. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

10. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-20

Title of Invention: Integral Weakness Consensus

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Integral Four-Quadrant Mapper, Controlling-Weakness Identification Engine, Eight-Model Consensus Scoring Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Integral Four-Quadrant + Controlling-Weakness Identification + Eight-Model Consensus Scoring to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Integral Four-Quadrant Mapper:** a deterministic structural mapping that converts a 32-dimension measurement into a four-quadrant integral representation without human interpretive variance.
- **Controlling-Weakness Identification Engine:** a constant-time deterministic diagnostic over a fixed-width vector that replaces unstructured narrative assessment with a reproducible, auditable computation.

- **Eight-Model Consensus Scoring Engine:** a multi-model scoring architecture that improves measurement reliability over any single model by combining a trimmed-mean outlier guard with per-model, per-dimension calibration weights accumulated from real scoring history — a specific, non-generic weighting data structure, not a mental process.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Integral Four-Quadrant Mapper. This engine maps the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context. *Implementation:* `server/patents/combinations/index.ts` (integral engine).

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* `server/scoring/controllingWeakness.ts`.

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* `server/scoring/consensus.ts`; `server/patents/calibrationBus.ts`; `server/patents/computeFabric.ts`.

Combinator. The combination executes as the `integralWeaknessConsensus` combinator (`server/patents/combinations/index.ts`), registered in the sister-combination registry (`server/patents/combinations/registry.ts`) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Integral Four-Quadrant Mapper:** Deterministic quadrant assignment of all 32 lines across interior/exterior x individual/collective; the mapping table is versioned with the rubric.

- **Controlling-Weakness Identification Engine:** Deterministic argmin over the fixed 32-dimension vector; constraint impact = $\text{clamp}(\text{profile mean minus minimum}, 0..1)$; measured cost 0.11 microseconds per operation.

- **Eight-Model Consensus Scoring Engine:** Panel of eight models; trimmed-mean guard active whenever four or more models contribute (single highest and single lowest raw scores excluded per line); calibration weight $w = 1/(\text{meanAbsErr} + 0.02)$ with an error floor preventing weight explosion; equal weight 1.0 until a model accumulates five observations on a line; confidence = $0.5 \times \text{mean model confidence} + 0.5 \times \text{inter-model agreement}$.

Measured Performance

- Controlling-weakness argmin over the 32-line vector: 0.11 microseconds per operation (8,745,295 ops/sec).

- Full 8-model x 32-line consensus combine: 0.11 ms (9,440 combines/sec).
- Panel fan-out over the compute fabric: 7.9x parallel speedup over serial execution (software backend; hardware co-processor figures are not claimed).

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Narrative integral-framework application: replaced by a deterministic, versioned mapping.
- Composite/average scoring instruments: the claimed method elevates the minimum, not the mean, following limiting-factor theory (Liebig; O-ring), and quantifies constraint impact.
- Single-model AI scoring and unweighted ensemble averaging: neither maintains per-model, per-dimension calibration history with a cold-start guarantee and a retained trim guard.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) mapping the 32-line profile onto the four quadrants of the AQAL integral framework (interior/exterior crossed with individual/collective), producing a structured integral view that situates each measured line in its quadrant context; (b) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (c) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein panel fan-out executes over a typed parallel compute fabric with bounded concurrency, a hardware co-processor backend being substitutable behind the same interface without alteration of the method.
6. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.
7. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.
8. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
9. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
10. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
11. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
12. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-27

Title of Invention: Voice-Driven Cognitive Coaching

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Prosodic Voice-Feature Extraction + Evidence-Mapped Protocol Coaching to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Prosodic Voice-Feature Extraction Engine:** an on-server digital-signal-processing pipeline whose energy-gated, lag-bounded pitch estimator eliminates the silent-frame false-pitch and subharmonic-bias failure modes of naive autocorrelation, verified by synthetic-tone tests — an improvement in the functioning of the audio-analysis computer itself.

- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Combinator. The combination executes as the voiceDrivenCognitiveCoaching combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Prosodic Voice-Feature Extraction Engine:** Decoded mono PCM analyzed in fixed 50 ms pitch frames; cumulative-mean-normalized difference threshold 0.15 with descent to the local minimum; maximum candidate lag bounded to half the frame length; silent-frame energy gate (mean energy per sample below 1e-4 returns no pitch), eliminating false pitch in silence.

- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.

Measured Performance

- Full voice pipeline on 30 s of audio: 5,270 ms — 6x faster than real time.

- Pitch recovery on synthetic tones: 110 Hz -> 110.3 Hz (0.3% error); 220 Hz -> 219.2 Hz (0.4%); 330 Hz -> 333.3 Hz (1.0%).

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Cloud speech-to-text vendors: prosody here is computed in-process on the operator's server under explicit on-page disclosure, with raw audio never transmitted to a third party.
- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQUAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, and bounds the maximum candidate lag to half the analysis-frame length, preventing false pitch in silence and subharmonic bias, and wherein the raw audio never leaves the server for the claimed analysis.
5. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.
6. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
7. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
8. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
9. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
10. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-28

Title of Invention: Transparent Cognitive Assessment

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Eight-Model Consensus Scoring Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Eight-Model Consensus Scoring + Transparency and Tamper-Evident Audit to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Eight-Model Consensus Scoring Engine:** a multi-model scoring architecture that improves measurement reliability over any single model by combining a trimmed-mean outlier guard with per-model, per-dimension calibration weights accumulated from real scoring history — a specific, non-generic weighting data structure, not a mental process.

- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the transparentCognitiveAssessment combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Eight-Model Consensus Scoring Engine:** Panel of eight models; trimmed-mean guard active whenever four or more models contribute (single highest and single lowest raw scores excluded per line); calibration weight $w = 1/(\text{meanAbsErr} + 0.02)$ with an error floor preventing weight explosion; equal weight 1.0 until a model accumulates five observations on a line; confidence = $0.5 \times \text{mean model confidence} + 0.5 \times \text{inter-model agreement}$.

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Full 8-model x 32-line consensus combine: 0.11 ms (9,440 combines/sec).

- Panel fan-out over the compute fabric: 7.9x parallel speedup over serial execution (software backend; hardware co-processor figures are not claimed).

- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Single-model AI scoring and unweighted ensemble averaging: neither maintains per-model, per-dimension calibration history with a cold-start guarantee and a retained trim guard.
- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQUAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (b) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein panel fan-out executes over a typed parallel compute fabric with bounded concurrency, a hardware co-processor backend being substitutable behind the same interface without alteration of the method.
6. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.
7. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
8. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
9. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
10. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
11. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-30

Title of Invention: Voice-Driven Cognitive Assessment

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Eight-Model Consensus Scoring Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Prosodic Voice-Feature Extraction + Eight-Model Consensus Scoring to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Prosodic Voice-Feature Extraction Engine:** an on-server digital-signal-processing pipeline whose energy-gated, lag-bounded pitch estimator eliminates the silent-frame false-pitch and subharmonic-bias failure modes of naive autocorrelation, verified by synthetic-tone tests — an improvement in the functioning of the audio-analysis computer itself.

- **Eight-Model Consensus Scoring Engine:** a multi-model scoring architecture that improves measurement reliability over any single model by combining a trimmed-mean outlier guard with per-model, per-dimension calibration weights accumulated from real scoring history — a specific, non-generic weighting data structure, not a mental process.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Combinator. The combination executes as the voiceDrivenCognitiveAssessment combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Prosodic Voice-Feature Extraction Engine:** Decoded mono PCM analyzed in fixed 50 ms pitch frames; cumulative-mean-normalized difference threshold 0.15 with descent to the local minimum; maximum candidate lag bounded to half the frame length; silent-frame energy gate (mean energy per sample below 1e-4 returns no pitch), eliminating false pitch in silence.

- **Eight-Model Consensus Scoring Engine:** Panel of eight models; trimmed-mean guard active whenever four or more models contribute (single highest and single lowest raw scores excluded per line); calibration weight $w = 1/(\text{meanAbsErr} + 0.02)$ with an error floor preventing weight explosion; equal weight 1.0 until a model accumulates five observations on a line; confidence = $0.5 \times \text{mean model confidence} + 0.5 \times \text{inter-model agreement}$.

Measured Performance

- Full voice pipeline on 30 s of audio: 5,270 ms — 6x faster than real time.

- Pitch recovery on synthetic tones: 110 Hz -> 110.3 Hz (0.3% error); 220 Hz -> 219.2 Hz (0.4%); 330 Hz -> 333.3 Hz (1.0%).

- Full 8-model x 32-line consensus combine: 0.11 ms (9,440 combines/sec).

- Panel fan-out over the compute fabric: 7.9x parallel speedup over serial execution (software backend; hardware co-processor figures are not claimed).

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Cloud speech-to-text vendors: prosody here is computed in-process on the operator's server under explicit on-page disclosure, with raw audio never transmitted to a third party.

- Single-model AI scoring and unweighted ensemble averaging: neither maintains per-model, per-dimension calibration history with a cold-start guarantee and a retained trim guard.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQUAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (c) combining the outputs of steps (a)–(b) into a single emergent result not produced by any component engine alone; and (d) committing the emergent result, together with a disclosure of every input that produced it, to an append-only

hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein panel fan-out executes over a typed parallel compute fabric with bounded concurrency, a hardware co-processor backend being substitutable behind the same interface without alteration of the method.
6. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, and bounds the maximum candidate lag to half the analysis-frame length, preventing false pitch in silence and subharmonic bias, and wherein the raw audio never leaves the server for the claimed analysis.
7. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.
8. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
9. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
10. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
11. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
12. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-34

Title of Invention: Cognitive-Financial Development Tracker

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Developmental Stage-Band Estimator, Controlling-Weakness Identification Engine, Evidence-Mapped Protocol Coaching Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Developmental Stage-Band + Controlling-Weakness Identification + Evidence-Mapped Protocol Coaching to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Developmental Stage-Band Estimator:** versioned-rubric banding that makes developmental staging reproducible: the same inputs under the same frozen rubric version always produce the same band.
- **Controlling-Weakness Identification Engine:** a constant-time deterministic diagnostic over a fixed-width vector that replaces unstructured narrative assessment with a reproducible, auditable computation.

- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Developmental Stage-Band Estimator. This engine places the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically. *Implementation:* server/platform/stageFramework.ts; references/NineStageRubric.pdf.

Controlling-Weakness Identification Engine. This engine deterministically identifies the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average. *Implementation:* server/scoring/controllingWeakness.ts.

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Combinator. The combination executes as the cognitiveFinancialDevelopmentTracker combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Developmental Stage-Band Estimator:** Stage banding computed under a fixed, versioned nine-stage rubric frozen per norming snapshot, so identical inputs reproduce identical bands at any later date.

- **Controlling-Weakness Identification Engine:** Deterministic argmin over the fixed 32-dimension vector; constraint impact = clamp(profile mean minus minimum, 0..1); measured cost 0.11 microseconds per operation.

- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.

Measured Performance

- Controlling-weakness argmin over the 32-line vector: 0.11 microseconds per operation (8,745,295 ops/sec).

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment

numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Practitioner-scored developmental staging: replaced by a frozen, versioned rubric computation that is reproducible and auditable.
- Composite/average scoring instruments: the claimed method elevates the minimum, not the mean, following limiting-factor theory (Liebig; O-ring), and quantifies constraint impact.
- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) placing the member's composite profile into a developmental stage band under a fixed, versioned nine-stage rubric, so stage context is computed reproducibly from the same frozen rubric for every member rather than impressionistically; (b) deterministically identifying the single line that most constrains the member's profile by an argmin over the 32-dimension score vector and quantifies its constraint impact as the distance of the weakest line below the profile mean, on the principle that a profile's outcome is capped by its scarcest input rather than its average; (c) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein the identified weakest dimension is reported with a constraint impact computed as the distance of the weakest dimension below the profile mean.
5. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.
6. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
7. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
8. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.
9. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
10. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

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DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-36

Title of Invention: Voice-Driven Cognitive Coaching with Transparency

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Evidence-Mapped Protocol Coaching Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Prosodic Voice-Feature Extraction + Evidence-Mapped Protocol Coaching + Transparency and Tamper-Evident Audit to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Prosodic Voice-Feature Extraction Engine:** an on-server digital-signal-processing pipeline whose energy-gated, lag-bounded pitch estimator eliminates the silent-frame false-pitch and subharmonic-bias failure modes of naive autocorrelation, verified by synthetic-tone tests — an improvement in the functioning of the audio-analysis computer

itself.

- **Evidence-Mapped Protocol Coaching Engine:** a recommendation structure in which every machine recommendation is bound at the data layer to its published evidence, making the recommendation chain mechanically traceable.

- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Evidence-Mapped Protocol Coaching Engine. This engine maps each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source. *Implementation:* shared/therapyLineMap.ts; server/patents/weaknessInterventions.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the voiceCognitiveCoachingTransparent combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Prosodic Voice-Feature Extraction Engine:** Decoded mono PCM analyzed in fixed 50 ms pitch frames; cumulative-mean-normalized difference threshold 0.15 with descent to the local minimum; maximum candidate lag bounded to half the frame length; silent-frame energy gate (mean energy per sample below 1e-4 returns no pitch), eliminating false pitch in silence.

- **Evidence-Mapped Protocol Coaching Engine:** Protocol-to-line mappings ranked PRIMARY/SECONDARY/TERTIARY; every mapping carries the published citation, the capacity the study shows the protocol develops, and the study's finding; recommendations are traceable score-to-protocol-to-source.

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Full voice pipeline on 30 s of audio: 5,270 ms — 6x faster than real time.
- Pitch recovery on synthetic tones: 110 Hz -> 110.3 Hz (0.3% error); 220 Hz -> 219.2 Hz (0.4%); 330 Hz -> 333.3 Hz (1.0%).
- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Cloud speech-to-text vendors: prosody here is computed in-process on the operator's server under explicit on-page disclosure, with raw audio never transmitted to a third party.
- Recommendation engines with opaque provenance: here the citation, capacity, and finding travel with every mapping at the schema level.
- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) extracting, in-process on the

server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) mapping each identified development target to specific capacity-development protocols drawn from an evidence-carrying library in which every protocol-to-line mapping carries a published citation, the capacity the literature shows the protocol develops, and the study's finding, so the recommendation chain is traceable from score to protocol to source; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.

2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.

3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.

4. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, and bounds the maximum candidate lag to half the analysis-frame length, preventing false pitch in silence and subharmonic bias, and wherein the raw audio never leaves the server for the claimed analysis.

5. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval [0,1], a violating write being rejected as an error rather than stored.

6. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.

7. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.

8. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a confidence threshold, and floors only rise.

9. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

10. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, measured technical improvement): **9.8**
- Enablement and written description (§112: operating parameters, module paths, measured behavior): **9.8**
- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in production; combinator invoked on every scoring run.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. This draft was prepared with AI assistance and is not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — AQAL-C-38

Title of Invention: Voice-Driven Cognitive Assessment with Transparency

Series: Series A — JoinAQAL Sister Combination (implemented; reduction to practice)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Cross-Reference to Related Disclosures

This combination builds on the AQAL core engine family (eight-model consensus scoring; prosodic voice analysis; controlling-weakness identification; verified achievement floors; versioned norming; research provenance) and the five shared components (append-only audit ledger; calibration-weighted consensus bus; parallel compute fabric; dimension isolation; voice-feature extraction), each separately documented in the company's technical disclosures.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessments.

Background

Single-instrument cognitive assessments produce point estimates with no audit trail, no verified lower bounds, and no mechanical link from a measured deficiency to the specific, evidence-supported interventions that address it.

Summary of the Invention

The invention combines the following engines into a single coordinated pipeline: Prosodic Voice-Feature Extraction Engine, Eight-Model Consensus Scoring Engine, Transparency and Tamper-Evident Audit Engine. The combination produces an emergent capability that no component produces alone, and every emergent result is disclosed to the member and committed to a tamper-evident hash-chained ledger. Under the KSR v. Teleflex framework, the claimed value rests on the combination producing synergistic effects a person of ordinary skill would not predict from the components individually — a position patent counsel must evaluate against prior art before filing.

Abstract

A computer-implemented system combining Prosodic Voice-Feature Extraction + Eight-Model Consensus Scoring + Transparency and Tamper-Evident Audit to produce an emergent, member-disclosed result committed to an append-only hash-chained audit ledger whose validity any party can verify by recomputation. The combination improves the reliability, reproducibility, and auditability of machine-generated assessment beyond any component operating alone.

Technical Improvements Over Prior Systems (§101 Positioning)

ELIGIBILITY POSITIONING (35 U.S.C. §101 / Alice). The claims are directed not to an abstract idea of assessment but to specific technical improvements in the functioning of the computing system that performs it, implemented through particular, named data structures and algorithms with measured performance characteristics:

- **Prosodic Voice-Feature Extraction Engine:** an on-server digital-signal-processing pipeline whose energy-gated, lag-bounded pitch estimator eliminates the silent-frame false-pitch and subharmonic-bias failure modes of naive autocorrelation, verified by synthetic-tone tests — an improvement in the functioning of the audio-analysis computer

itself.

- **Eight-Model Consensus Scoring Engine:** a multi-model scoring architecture that improves measurement reliability over any single model by combining a trimmed-mean outlier guard with per-model, per-dimension calibration weights accumulated from real scoring history — a specific, non-generic weighting data structure, not a mental process.

- **Transparency and Tamper-Evident Audit Engine:** an append-only hash-chained audit structure in which any mutation of history is detectable by recomputation — tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator.

The claims recite these particular structures — the hash-chain commitment format, the calibration-weight formula and cold start, the energy-gated lag-bounded pitch estimator, the fixed-width vector contract, frozen norming versions, and frozen isolation views — rather than a result-only aspiration, and each is tied to measured behavior below.

Detailed Description

Prosodic Voice-Feature Extraction Engine. This engine extracts, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis. *Implementation:* server/patents/voiceFeatures.ts; voice_features table (migration 0030).

Eight-Model Consensus Scoring Engine. This engine scores each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence. *Implementation:* server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Transparency and Tamper-Evident Audit Engine. This engine discloses to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring. *Implementation:* server/patents/ledger.ts; /api/ledger/verify; /api/ledger/head.

Combinator. The combination executes as the voiceCognitiveAssessmentTransparent combinator (server/patents/combinations/index.ts), registered in the sister-combination registry (server/patents/combinations/registry.ts) and invoked in the production scoring pass over the real outputs of each scoring run.

Shared substrate. The combination inherits: the append-only audit ledger (each combined output is committed to a hash-chained ledger whose validity is publicly verifiable end to end); the calibration-weighted consensus bus (per-model, per-dimension weights derived from accumulated distance to the settled panel consensus, with an equal-weight cold start); dimension isolation (per-dimension frozen views make cross-dimension reads impossible by construction during scoring); verified achievement floors (a per-line floor that only rises, set by the repeated-demonstration rule (the minimum of two independently demonstrated scores at adequate confidence)); research provenance (each of the 32 lines is mapped in a database table to the named research tradition and scholars grounding it, with DOI fields left null rather than fabricated).

Operating Parameters (Enablement Detail)

- **Prosodic Voice-Feature Extraction Engine:** Decoded mono PCM analyzed in fixed 50 ms pitch frames; cumulative-mean-normalized difference threshold 0.15 with descent to the local minimum; maximum candidate lag

bounded to half the frame length; silent-frame energy gate (mean energy per sample below $1e-4$ returns no pitch), eliminating false pitch in silence.

- **Eight-Model Consensus Scoring Engine:** Panel of eight models; trimmed-mean guard active whenever four or more models contribute (single highest and single lowest raw scores excluded per line); calibration weight $w = 1/(\text{meanAbsErr} + 0.02)$ with an error floor preventing weight explosion; equal weight 1.0 until a model accumulates five observations on a line; confidence = $0.5 \times \text{mean model confidence} + 0.5 \times \text{inter-model agreement}$.

- **Transparency and Tamper-Evident Audit Engine:** Entry hash = SHA-256 over (previous hash | entry kind | canonical JSON with recursively sorted keys); genesis value is the 64-zero string; a unique index on the hash column rejects duplicate commitments; verification recomputes every link.

Measured Performance

- Full voice pipeline on 30 s of audio: 5,270 ms — 6x faster than real time.
- Pitch recovery on synthetic tones: 110 Hz -> 110.3 Hz (0.3% error); 220 Hz -> 219.2 Hz (0.4%); 330 Hz -> 333.3 Hz (1.0%).
- Full 8-model x 32-line consensus combine: 0.11 ms (9,440 combines/sec).
- Panel fan-out over the compute fabric: 7.9x parallel speedup over serial execution (software backend; hardware co-processor figures are not claimed).
- Hash-chain append throughput: 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

All figures measured by the repository's benchmark harness (scripts/patentBenchmarks.test.ts; docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22. They are software-embodiment numbers and are not presented as hardware co-processor performance.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrasts are provided to focus counsel's prior-art search; they are the applicant's engineering understanding, not an assertion that no prior art exists.

- Cloud speech-to-text vendors: prosody here is computed in-process on the operator's server under explicit on-page disclosure, with raw audio never transmitted to a third party.
- Single-model AI scoring and unweighted ensemble averaging: neither maintains per-model, per-dimension calibration history with a cold-start guarantee and a retained trim guard.
- Conventional mutable audit logs and W3C PROV-style dataset-level provenance: the claimed chain is append-only, self-verifying by hash recomputation, publicly checkable, and anchored per-result rather than per-dataset.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). The claimed combination is IMPLEMENTED AND OPERATING as a software embodiment in the JoinAQAL codebase: the combinator function named below executes in production immediately after each scoring run, over the real outputs of that run, and commits its emergent result to the tamper-evident audit ledger. This constitutes reduction to practice of the software embodiment. Hardware embodiments named in related specifications (FPGA co-processing, secure-enclave execution, HSM-signed ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Every output of the claimed combination is produced under the platform's standing gates: outputs are educational and non-clinical; all inputs to each emergent result are disclosed to the member; no medical, diagnostic, or treatment claims are made; and voice-derived analysis, where present, is

disclosed on-page before capture, with audio processed in-process on the server and never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a networked assessment system by producing a tamper-evident emergent cognitive-development result, the method comprising: (a) extracting, in-process on the server and only after on-page disclosure to the member, a prosodic feature vector from recorded spoken answers: fundamental-frequency estimation by a cumulative-mean-normalized-difference (YIN-family) method over fixed 50-millisecond analysis frames with a silent-frame energy gate to suppress false pitch in silence, plus speaking rate, pause count and duration statistics, hesitation frequency, jitter, shimmer, spectral centroid, and RMS energy, persisted per assessment; the raw audio never leaves the server for this analysis; (b) scoring each of the 32 intelligence lines by fanning the member's responses to a panel of eight independent AI models in parallel over a typed compute fabric, discarding the single highest and single lowest raw score per line (trimmed-mean outlier guard), and combining the survivors under the calibration-weighted consensus bus, in which each model's weight on each line is derived from its accumulated historical distance from the settled panel consensus (equal-weight cold start until a minimum sample count is reached), yielding a per-line score and an agreement-derived confidence; (c) disclosing to the member every input that produced the emergent result and commits the result to an append-only, hash-chained audit ledger in which each entry's hash commits to the previous entry's hash plus the canonical serialization of its own payload, so any after-the-fact edit, deletion, or reordering of history breaks verification from that point forward; chain validity is publicly checkable and the chain head is exportable for external anchoring; (d) combining the outputs of steps (a)–(c) into a single emergent result not produced by any component engine alone; and (e) committing the emergent result, together with a disclosure of every input that produced it, to an append-only hash-chained audit ledger.
2. The method of claim 1, wherein the audit ledger's validity is verifiable end to end by recomputing every link of the hash chain, and the current chain head is exportable for external anchoring by hardware signature, trusted timestamp, or public chain.
3. The method of claim 1, wherein every output is produced under standing safety gates comprising: educational and non-clinical outputs only; disclosure of all inputs to the member; and the making of no medical, diagnostic, or treatment claims.
4. The method of claim 1, wherein each panel model's per-dimension weight is derived from its accumulated distance to the settled panel consensus, with equal weighting until a minimum observation count is reached, and a trimmed-mean outlier guard retained beneath the weights.
5. The method of claim 1, wherein panel fan-out executes over a typed parallel compute fabric with bounded concurrency, a hardware co-processor backend being substitutable behind the same interface without alteration of the method.
6. The method of claim 1, wherein fundamental-frequency estimation applies a silent-frame energy gate that returns no pitch for frames below an energy threshold, and bounds the maximum candidate lag to half the analysis-frame length, preventing false pitch in silence and subharmonic bias, and wherein the raw audio never leaves the server for the claimed analysis.
7. The method of claim 1, wherein a validation layer enforces a fixed-width vector contract of exactly 32 dimensions with unique indices and scores on the interval $[0,1]$, a violating write being rejected as an error rather than stored.
8. The method of claim 1, wherein each stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical population-rarity mapping at any later time, and a public changelog enumerates every version.
9. The method of claim 1, wherein per-dimension computation executes over frozen isolation views that prevent cross-dimension reads by construction.
10. The method of claim 1, wherein per-line verified floors are maintained by a repeated-demonstration rule under which a floor is set to the minimum of two scores demonstrated on separate completed assessments at or above a

confidence threshold, and floors only rise.

11. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

12. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. This is NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

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